

2-Hour Developer Assessment: AI-Powered Personal Finance Tracker

Problem Statement

Build a **functional MVP personal finance tracker** with budget planning in **2 hours** using AI tools effectively. Submit as a complete GitHub repository with clear documentation. Make functional, strategic & technical decisions and make sure the decisions can be justified with proper reasoning.

Time-Boxed Challenge: 2 Hours Maximum

Hour 1: Setup & Core Features

- Project scaffolding and basic architecture
- User authentication (simplified) - Choose google auth if it saves time
- Income/Expense tracking
- Basic budget creation

Hour 2: Polish & Documentation

- Budget vs actual comparison
- Data visualization components
- Responsive design touches
- Documentation and GitHub repo preparation

MVP Feature Selection Matrix

MUST HAVE (Core MVP)

Pick ALL of these - justify if you skip any:

- 1. Transaction Management**
 - Add income/expense transactions
 - View transaction list with date, amount, description
 - Basic categorization (Food, Transport, Entertainment, etc.)
 - Edit/delete transactions
- 2. Budget Creation**
 - Create monthly budgets by category
 - Set spending limits for each category
 - Simple form for budget setup
- 3. Basic Dashboard**
 - Total income vs expenses for current month

- Remaining budget overview
- Simple balance calculation

SHOULD HAVE (Choose 2-3 features)

Justify your selection based on business impact vs. implementation effort:

4. Budget vs Actual Comparison

- Show spent vs budgeted amount per category
- Visual indicators (green/yellow/red) for budget status
- Percentage calculations and warnings

5. Data Visualization

- Pie chart for expense breakdown by category
- Bar chart for monthly spending trends
- Simple charts using Chart.js or similar

6. Recurring Transactions

- Mark transactions as recurring (monthly/weekly)
- Auto-generate recurring entries
- Manage recurring transaction templates

7. Transaction Search & Filter

- Search transactions by description
- Filter by category, date range, amount
- Sort by date, amount, category

8. Savings Goals Tracker

- Set savings goals with target amounts
- Track progress toward goals
- Visual progress indicators

9. Account Management

- Multiple account support (Checking, Savings, Credit Card)
- Account balances and transfers
- Transaction assignment to accounts

10. Monthly Budget Templates

- Save budget templates for reuse
- Copy previous month's budget
- Quick budget setup options

COULD HAVE (Bonus - if time permits)

Only attempt if core features are solid:

11. Expense Analytics

- Monthly comparison charts

- Spending pattern insights
- Category-wise trend analysis

12. Export Functionality

- Export transactions to CSV
- Export monthly reports
- Print-friendly budget summaries

13. Bill Reminders

- Set up bill due dates
- Payment reminders and notifications
- Mark bills as paid

14. Mobile-Optimized Input

- Quick expense entry with preset amounts
- Camera integration for receipt scanning (mock)
- Touch-friendly mobile interface

15. Budget Alerts

- Notifications when approaching budget limits
- Overspend warnings
- Monthly budget reset notifications

16. Income vs Expense Reports

- Monthly income/expense summaries
- Year-to-date comparisons
- Profit/loss style reporting

17. Dark Mode & Themes

- Toggle between light/dark themes
- Color-coded categories
- Accessibility improvements

18. Data Import/Export

- Import from bank CSV files
- Backup/restore functionality
- Data migration tools